数理方程

第11周
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球函数：

Legendre 方程
- 微分表示
- 广义傅里叶级数
- 生成函数

连带 Legendre 方程
- 微分表示
- 广义傅里叶级数
- 球谐函数

Adrien-Marie Legendre



Watercolor caricature by Julien-Léopold Boilly (see § Mistaken portrait), the only known portrait of Legendre[2]

Legendre方程

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{S(\theta,\phi)}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2 S(\theta,\phi)}{\partial\phi^2}+\lambda S(\theta,\phi)=0$$

$S(\theta,\phi)=\Theta(\theta)\Phi(\phi)$ ⇓ $\left[\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\lambda\sin^2\theta\,\Theta\right]\Phi=-\frac{d^2\Phi}{d\phi^2}\Theta$

⇓

$$\Rightarrow\left[\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\lambda\sin^2\theta\,\Theta\right]\frac{1}{\Theta}=-\frac{d^2\Phi}{d\phi^2}\frac{1}{\Phi}=\mu$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{\mu}{\sin^2\theta}\right)\Theta=0,$$

$\sin^2\theta\cdot\frac{1}{\sin\theta}$

⇓

$x=\cos\theta,\ -1\le x\le 1$

$$\frac{d}{dx}=\frac{d\theta}{dx}\frac{d}{d\theta}=-\frac{1}{\sin\theta}\frac{d}{d\theta}$$

本征函数

$$\frac{d}{dx}\left[(1-x^2)\frac{d\Theta}{dx}\right]+\left(\lambda-\frac{\mu}{1-x^2}\right)\Theta=0$$

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right]+\left(\lambda-\frac{\mu}{1-x^2}\right)y=0$$

$$(1-x^2)y''-2xy'+\left(\lambda-\frac{m^2}{1-x^2}\right)y=0$$

$$\frac{d^2\Phi}{d\phi^2}+\mu\Phi=0$$

$\Phi(0)=\Phi(2\pi)$

$\Phi'(0)=\Phi'(2\pi)$

本征值 $\mu_m=m^2$, $m=0,1,2\cdots$

本征函数 $\Phi_{m1}=\sin m\phi$, $\Phi_{m2}=\cos m\phi$ $m=1,2\cdots$

$\Phi_0=1$

$m=0$: Legendre方程

$m=1,2,3\cdots$ 连带Legendre associated

$m=0$; $\Phi(\phi)=\text{const}$. 绕z轴对称.

$m\neq 0$ 无轴对称.

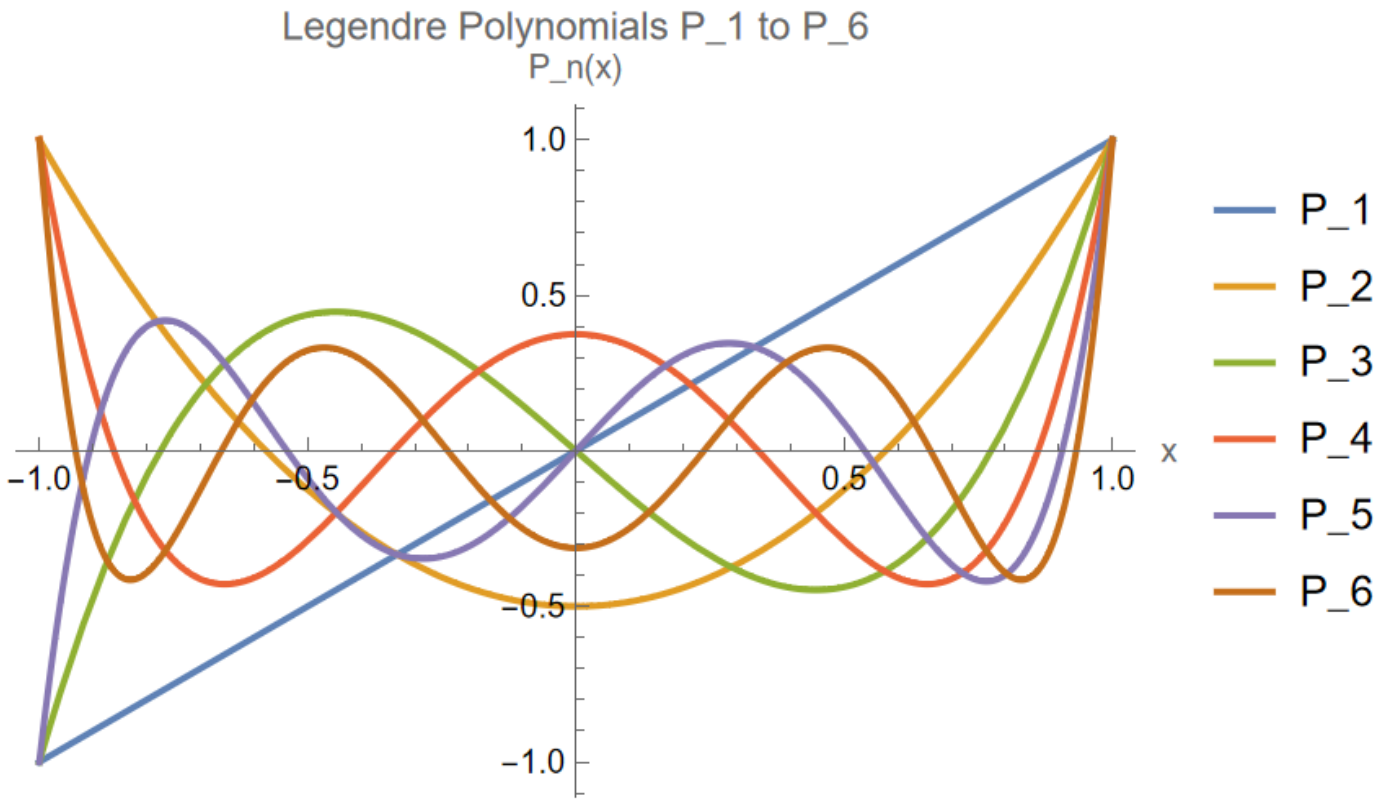
$m=0$ $\quad(1-x^2)y''-2xy'+\lambda y=0;\quad y(\pm1)<\infty$

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right]+\lambda y=0$$

常点

$\Rightarrow$ $\lambda_l = l(l+1)$ $l = 0, 1, 2 \cdots$

$P_l(x)$



(1) 微分表示. (Rodrigues formula 1816年)

$$P_l(x) = \frac{1}{2^l l!}\left[(x^2-1)^l\right]^{(l)}$$ ↑ l阶导数

例: $P_1(x) = \frac{1}{2}\frac{d}{dx}(x^2-1) = x$

$P_2(x) = \frac{1}{8}\frac{d^2}{dx^2}\left[x^4-2x^2+1\right] = \frac{1}{2}(3x^2-1)$

推导:

记 $y=(x^2-1)^l \Rightarrow y'=2lx(x^2-1)^{l-1} \Rightarrow (x^2-1)y'=2lx(x^2-1)^l$

$=2lxy$

Leibniz rule $(uv)^{(n)}=\sum_{k=0}^{n} C_n^k u^{(n-k)}v^{(k)}$

$=u^{(n)}v+nu^{(n-1)}v'+\frac{n(n-1)}{2!}u^{(n-2)}v''+\cdots$

$(uv)''=u''v+2u'v'+uv''$

求 $l+1$ 阶导数

$(x^2-1)y^{(l+2)}+(l+1)y^{(l+1)}2x+\frac{l(l+1)}{2}y^{(l)}\cdot 2=2lxy^{(l+1)}+2l(l+1)y^{(l)}$

$\Rightarrow (1-x^2)y^{(l+2)}-2xy^{(l+1)}+l(l+1)y^{(l)}=0$

$\Rightarrow y^{(l)}$ 满足 Legendre 方程

$\Rightarrow P_l(x)=\frac{1}{y^{(l)}(1)}y^{(l)}(x)$ $\quad P_l(1)=1$

$y^{(l)}(1)=[(x^2-1)^l]^{(l)}\Big|_{x=1}=[(x-1)^l(x+1)^l]^{(l)}\Big|_{x=1}$

$=\underbrace{[(x-1)^l]^{(l)}(x+1)^l\Big|_{x=1}}_{\neq 0}+l\underbrace{[(x-1)^l]^{(l-1)}[(x+1)^l]'\Big|_{x=1}}_{=0}+\cdots$

$=l!2^l$

$\Rightarrow P_l(x)=\frac{1}{2^l l!}[(x^2-1)^l]^{(l)}$

例: $\int_{-1}^{1} x^k P_l(x)\,dx$ $\quad k,l=1,2,3,\cdots$

$P_l(x)=(-1)^l P_l(-x)$ $\quad k\pm l=$奇数 $\Rightarrow$ 积分$=0$

$k\pm l=$偶数

$$\int_{-1}^{1} x^k P_l(x)\,dx=\frac{1}{2^l l!}\int_{-1}^{1} x^k\left[(x^2-1)^l\right]^{(l)}dx$$

$$=\frac{1}{2^l l!}x^k\left[(x^2-1)^l\right]^{(l-1)}\Big|_{-1}^{1}-\frac{1}{2^l l!}\int_{-1}^{1}(x^k)'\left[(x^2-1)^l\right]^{(l-1)}dx$$

$[(x-1)^l(x+1)^l]^{(l-1)}=[(x-1)^l]^{(l-1)}(x+1)^l+l[(x-1)^l]^{(l-2)}[(x+1)^l]'+\cdots$
$=0$

多次分部积分

$$=(-1)^l\frac{1}{2^l l!}\int_{-1}^{1}(x^k)^{(l)}(x^2-1)^l dx=\frac{1}{2^l l!}\int_{-1}^{1}(x^k)^{(l)}(1-x^2)^l dx$$

$k<l\Rightarrow (x^k)^{(l)}=0\Rightarrow 0$

$k\geq l$ 记 $k=l+2n$ $(n\geq 0)$

$$\Rightarrow\int_{-1}^{1}x^{l+2n}P_l(x)dx=\frac{1}{2^l l!}\int_{-1}^{1}(x^{l+2n})^{(l)}(1-x^2)^l dx$$

$(l+2n)(l+2n-1)\cdots(1+2n)x^{2n}=\frac{(l+2n)!}{(2n)!}x^{2n}$

$$=\frac{1}{2^l l!}\frac{(l+2n)!}{(2n)!}\int_{-1}^{1}x^{2n}(1-x^2)^l dx$$

$\Downarrow\ x^2=t$ $\quad 2x\,dx=dt$, $\int_{-1}^{1}=2\int_{0}^{1}$

$$=\frac{1}{2^l l!}\frac{(l+2n)!}{(2n)!}\int_{0}^{1}t^{n-\frac{1}{2}}(1-t)^l dt$$

数树 P_{113}

$$B(p,q)=\int_0^1 t^{p-1}(1-t)^{q-1}dt=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$ $\quad p=n+\frac{1}{2}$, $q=l+1$

$\Gamma(x)=\int_0^\infty e^{-t}t^{x-1}dt$

$\Gamma(x+1)=\int_0^\infty t^x d(-e^{-t})=-t^x e^{-t}\Big|_0^\infty+\int_0^\infty x t^{x-1}e^{-t}dt=x\Gamma(x)$

$$= \frac{1}{2^l l!}\frac{(l+2n)!}{(2n)!}\frac{\Gamma(n+\frac12)\Gamma(l+1)}{\Gamma(n+l+\frac32)} = 2^{l+1}\frac{(l+2n)!}{n!}\frac{(l+n)!}{(2l+2n+1)!}$$

多次利用 $\Gamma(x+1)=x\Gamma(x)\Rightarrow \Gamma(n+l+\frac32)=(n+l+\frac12)(n+l-\frac12)\cdots(1+\frac12)\frac12\Gamma(\frac12)$

$$= \frac{1}{2^{n+l+1}}(2n+2l+1)(2n+2l-1)\cdots(0+1) = \frac{1}{2^{2(n+l)+1}}\frac{(2n+2l+1)!}{(n+l)!}\Gamma(\tfrac12),$$

同理, $\frac{\Gamma(n+\frac12)}{(2n)!} = \frac{1}{2^{2n}n!}\Gamma(\frac12)$.

$n=0$ ($k=l$):

$$\int_{-1}^{1}x^l P_l(x)dx = 2^{l+1}\frac{(l!)^2}{(2l+1)!}$$

设: $I_l = \int_0^1 P_l(x)dx$

$l=2n$ ($n=1,2,3\cdots$) $\Rightarrow I_{2n} = \frac12\int_{-1}^{1}P_{2n}(x)dx = 0$

$P_{2n}(x)$ 与 $P_0(x)$, 正交

$l=2n+1$ ($n=0,1,2$)

$$\Rightarrow I_{2n+1} = \frac{1}{2^{2n+1}(2n+1)!}\int_0^1\left[(x^2-1)^{2n+1}\right]^{(2n+1)}dx$$

$$= \frac{1}{2^{2n+1}(2n+1)!}\left[(x^2-1)^{2n+1}\right]^{(2n)}\Big|_0^1$$

由 Leibniz rule, $x=1$ 时值为 0

$x^2-1=(x-1)(x+1)$.

$$= -\frac{1}{2^{2n+1}}\frac{1}{(2n+1)!}\left[(x^2-1)^{2n+1}\right]^{(2n)}\Big|_{x=0}$$

$$= -\frac{1}{2^{2n+1}(2n+1)!}\left[C_{2n+1}^{n}(x^2)^n(-1)^{n+1}\right]^{(2n)}\Big|_{x=0} = (-1)^n\frac{1}{2^{2n+1}}\frac{(2n+1)!}{(n+1)!n!}(2n)!$$

仅 x^{2n} 项有贡献

$$= (-1)^n\frac{(2n-1)!!}{(2n+2)!!}$$

双阶乘 $5!! = 5\cdot3\cdot1$, $6!! = 6\cdot4\cdot2$

$\frac{(2n)!}{n!2^n} = (2n-1)!!$

$\frac{1}{(n+1)!2^{n+1}} = \frac{1}{(2n+2)!!}$

(2) 正交性与广义傅里叶级数

$\int_{-1}^{1}P_k(x)P_l(x)dx = 0$ ($k\neq l$) $\quad \frac{d}{d\theta}\left(\sin\theta\frac{dy}{d\theta}\right)+\lambda\sin\theta y = 0$

$\Downarrow x=\cos\theta$

$\int_0^{\pi}P_k(\cos\theta)P_l(\cos\theta)\sin\theta\,d\theta = 0$, ($k\neq l$)

$$\|P_l\|^2 = \int_{-1}^{1} [P_l(x)]^2 dx = \frac{2}{2l+1}$$ ← 可由微分表示得出，由生成函数计算更简便

$$[-1,1]: \quad f(x) = \sum_{l=0}^{\infty} c_l P_l(x), \quad c_l = \frac{1}{\|P_l\|^2} \int_{-1}^{1} f(x) P_l(x) dx$$

$$= \frac{2l+1}{2} \int_{-1}^{1} f(x) P_l(x) dx$$

有时也可走近路：$x^2 = \frac{2}{3}\left[\frac{1}{2}(3x^2-1)\right] + \frac{1}{3} = \frac{2}{3}P_2(x) + \frac{1}{3}P_0(x)$

$x^k = (\cdots)P_k + (\cdots)P_{k-1} + \cdots + (\cdots)P_0$，利用 $P_l(x)$ 表达式定系数

$$[0,\pi]: \quad f(\theta) = \sum_{l=0}^{\infty} c_l P_l(\cos\theta), \quad c_l = \frac{2l+1}{2} \int_0^{\pi} f(\theta) P_l(\cos\theta) \sin\theta \, d\theta$$

(3) 生成函数，母函数 generating function

例：Fibonacci $a_0 = a_1 = 1$,

1 1 2 3 5 8 13

$a_n = a_{n-1} + a_{n-2} \quad (n \geq 2)$

↓ 乘 t^n，对 n 求和

$$\sum_{n=2}^{\infty} a_n t^n = \sum_{n=2}^{\infty} (a_{n-1}t^n + a_{n-2}t^n)$$

本例与PDE无关，不作要求，仅供辅助理解，不感兴趣可跳过

生成函数

$$f(t) = a_0 + a_1 t + a_2 t^2 + \cdots$$

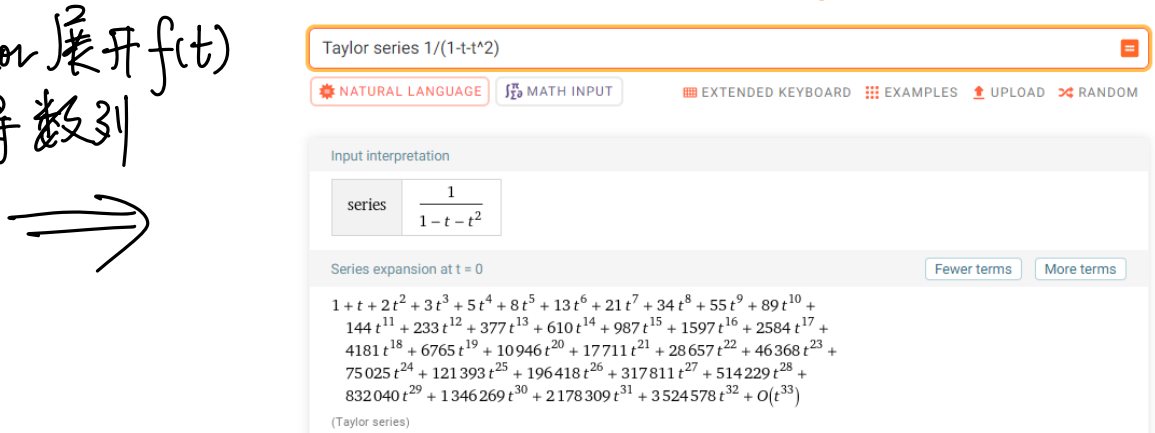
所有 a_n 打包在一起

$$f(t) - a_0 - a_1 t = t[f(t) - a_0] + t^2 f(t)$$

$$\Rightarrow (1 - t - t^2) f(t) = a_0 + a_1 t - a_0 t = 1$$

$$f(t) = \frac{1}{1 - t - t^2}$$

Taylor 展开 $f(t)$ 可得数列 ⇒



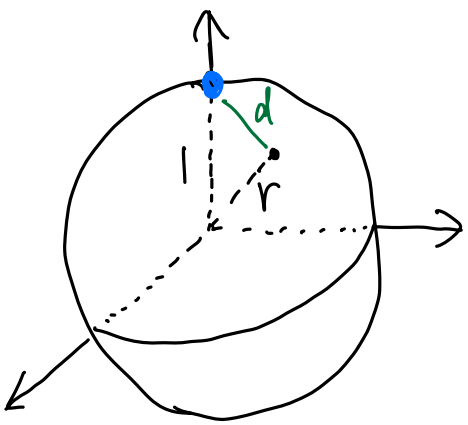
$$f(t)=\frac{1}{\left(\frac{\sqrt5+1}{2}+t\right)\left(\frac{\sqrt5-1}{2}-t\right)}=\frac{1}{\sqrt5}\left[\frac{1}{\frac{\sqrt5+1}{2}+t}+\frac{1}{\frac{\sqrt5-1}{2}-t}\right]$$

$$=\frac{1}{\sqrt5}\sum_{n=0}^{\infty}\left[\left(\frac{1+\sqrt5}{2}\right)^{n+1}-\left(\frac{1-\sqrt5}{2}\right)^{n+1}\right]t^n$$

$$\Rightarrow a_n=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{n+1}-\left(\frac{1-\sqrt5}{2}\right)^{n+1}\right]$$

生成函数：$\{a_n\}$的信息翻译为$f(t)$的信息，后者容易处理，可导出更多信息，然后再翻译为$\{a_n\}$的信息.

$$g(x,t)=\sum_{l=0}^{\infty}P_l(x)t^l=\frac{1}{\sqrt{1-2xt+t^2}}$$



北极$(x,y,z)=(0,0,1)$处放一点电荷，

(r,θ,ϕ) $r<1$ 单位球内

$$d=\sqrt{1-2r\cos\theta+r^2}$$

$$\Rightarrow \text{球内电势 } u=\frac{1}{d}=\frac{1}{\sqrt{1-2r\cos\theta+r^2}}$$

球内无电荷$\Rightarrow u$满足 $\nabla^2 u=\nabla^2\left(\frac{1}{d}\right)=0$ $r<1$

分离变量法$\Rightarrow u$的级数解，正是$\sum_{l=0}^{\infty}P_l(\cos\theta)r^l$！ 计算如下：

$$u=R(r)\Phi(\theta)\Rightarrow\begin{cases}\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Phi}{d\theta}\right)+\lambda\Phi=0\\ \frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\lambda R=0\end{cases}\Rightarrow\begin{cases}\lambda_l=l(l+1)\\ P_l(\cos\theta)\end{cases}$$

$l=0,1,2\cdots$

$r=e^t$, $\frac{d}{dt}=r\frac{d}{dr}$ $\Rightarrow$ $\frac{d^2R(t)}{dt^2}+\frac{dR(t)}{dt}-l(l+1)R(t)=0$

$\Rightarrow R=A_l e^{lt}+B_l e^{-(l+1)t}=A_l r^l+B_l\frac{1}{r^{l+1}}$

$\Rightarrow u=\frac{1}{d}=\sum_{l=0}^{\infty}(A_l r^l+B_l\frac{1}{r^{l+1}})P_l(\cos\theta)$, $r<1$

($\frac{1}{r^{l+1}}\to\infty$)

$=\sum_{l=0}^{\infty}A_l r^l P_l(\cos\theta)$

利用 $P_l(1)=1$ $\overset{\theta=0}{=\!=}$ $\sum_{l=0}^{\infty}A_l r^l$

$\theta=0\Rightarrow \frac{1}{d}=\frac{1}{1-r}$

$=1+r+r^2+r^3+\cdots$

$A_l=1$

$\Rightarrow \frac{1}{d}=\frac{1}{\sqrt{1-2r\cos\theta+r^2}}=\sum_{l=0}^{\infty}r^l P_l(\cos\theta)$

$\Downarrow$ $r\to t$, $\cos\theta\to x$

$g(x,t)=\frac{1}{\sqrt{1-2xt+t^2}}=\sum_{l=0}^{\infty}t^l P_l(x)$ $|t|<1$

生成函数

根据复变函数，$g(x,t)$有奇点 $t=x\pm\sqrt{1-x^2}i$

$\Rightarrow$ 收敛半径为 $|t|=1$

$|t|>1\Rightarrow g(x,t)=\frac{1}{\sqrt{1-2xt+t^2}}=\frac{1}{t}\frac{1}{\sqrt{1-2x\frac{1}{t}+(\frac{1}{t})^2}}$

$=\frac{1}{t}\sum_{l=0}^{\infty}(\frac{1}{t})^l P_l(x)=\sum_{l=0}^{\infty}\frac{1}{t^{l+1}}P_l(\cos\theta)$

应用①：$\|P_l\|^2$

$[g(x,t)]^2=\frac{1}{1-2xt+t^2}=[\sum_{l=0}^{\infty}P_l(x)t^l]^2$

$$\Rightarrow \int_{-1}^{1}\frac{dx}{1-2xt+t^2}=\sum_{l=0}^{\infty}t^{2l}\int_{-1}^{1}[P_l(x)]^2dx \quad (\text{利用正交性, } \int_{-1}^{1}P_lP_k\,dx=0,\ l\neq k)$$

$$\downarrow y=1-2xt+t^2,\ dy=-2t\,dx$$

$$\frac{1}{2t}\int_{(1-t)^2}^{(1+t)^2}\frac{dy}{y}=\frac{1}{t}\ln\frac{1+t}{1-t}=2\sum_{l=0}^{\infty}\frac{t^{2l}}{2l+1}\Rightarrow\int_{-1}^{1}[P_l(x)]^2=\frac{2}{2l+1}$$

应用②：递推关系

① $(l+1)xP_l(x)=(l+1)P_{l+1}(x)+lP_{l-1}(x)$ ← $g(x,t)$对t求导

② $P'_{l+1}(x)+P'_{l-1}(x)=2xP'_l(x)+P_l(x)$ ← $g(x,t)$对x求导

③ $P'_{l+1}(x)=xP'_l(x)+(l+1)P_l(x)$

④ $P'_{l-1}(x)=xP'_l(x)-lP_l(x)$

证①：$g(x,t)=(1-2xt+t^2)^{-\frac{1}{2}}=\sum_{l=0}^{\infty}P_l(x)t^l$.

$$\frac{\partial g}{\partial t}=\frac{x-t}{(1-2xt+t^2)^{3/2}}=\sum_{l=0}^{\infty}lP_l(x)t^{l-1}$$

$$\Downarrow \text{乘}\ 1-2xt+t^2$$

$$(x-t)\left(\sum_{l=0}^{\infty}P_l(x)t^l\right)=(1-2xt+t^2)\sum_{l=0}^{\infty}lP_l(x)t^{l-1}\xrightarrow{\text{比较}t\text{的系数}}①$$

应用：$\int_{-1}^{1}xP_k(x)P_l(x)\,dx,\quad k,l=1,2,\cdots$

↑ 利用递推关系化为P_{k+1}与P_{k-1}

连带Legendre方程 $m=1,2,3\cdots$

$$(1-x^2)y''-2xy'+\left(\lambda-\frac{m^2}{1-x^2}\right)y=0 \qquad y(\pm1)<\infty$$

(1) 微分表示

① $y(x)=(1-x^2)^{\frac{m}{2}}v(x)$

为何引入$(1-x^2)^{\frac{m}{2}}$? 在$x=1$附近求解y, $t=x-1$

$$\Rightarrow -t(t+2)y''-2(t+1)y'+\left[\lambda+\frac{m^2}{t(t+2)}\right]y=0$$

指标方程 $\rho^2=\frac{m^2}{4}$, $\rho=\pm\frac{m}{2}$ $\xrightarrow{y\text{不发散}}$ $\rho=\frac{m}{2}$ $\Rightarrow y=t^{\frac{m}{2}}(a_0+a_1t+\cdots)$

在$t=0$附近求级数解

$$y(x)=(x-1)^{\frac{m}{2}}[a_0+a_1(x-1)+\cdots]$$

$x=-1$附近求解: $y(x)=(x+1)^{\frac{m}{2}}[b_0+b_1(x-1)+\cdots]$,

$$y(x)=(1-x)^{\frac{m}{2}}(1+x)^{\frac{m}{2}}v(x)=(1-x^2)^{\frac{m}{2}}v(x)$$

为了消除$(x+1)^{\frac{m}{2}}$和$(x+1)^{\frac{m}{2}}$

代入原方程 $\Rightarrow (1-x^2)^{\frac{m}{2}+1}v''-2(m+1)x(1-x^2)^{\frac{m}{2}}v'+(1-x^2)^{\frac{m}{2}}[\lambda-m(m+1)]v=0$

$$\Rightarrow (1-x^2)v''-2(m+1)xv'+[\lambda-m(m+1)]v=0$$

② $v(x)$方程 $\longleftrightarrow$ Legendre方程

$P_l(x)$满足Legendre方程: $[(1-x^2)P_l']'+\lambda P_l=0$, $P_l(\pm1)<\infty$

$\lambda=l(l+1)$

求m次导数 $\Rightarrow$ $[(1-x^2)P_l']^{(m+1)}+\lambda P_l^{(m)}=0$

$$\Rightarrow (1-x^2)(P_l')^{(m+1)}+(m+1)(P_l')^{(m)}(-2x)+\frac{m}{2}(m+1)(P_l')^{(m-1)}(-2)+\lambda P_l^{(m)}=0$$

$$\Rightarrow (1-x^2)P_l^{(m+2)}-2(m+1)xP_l^{(m+1)}+[\lambda-m(m+1)]P_l^{(m)}=0$$

$P_l^{(m)}$与v满足同一方程.

$$\Rightarrow v(x)\propto[P_l(x)]^{(m)} \qquad v(x)=(-1)^m[P_l(x)]^{(m)}$$

习惯取比例为$(-1)^m$

$\Rightarrow$ 本征函数 $y_l(x) \equiv P_l^m(x) = (1-x^2)^{\frac{m}{2}} v = (-1)^m (1-x^2)^{\frac{m}{2}} [P_l(x)]^{(m)}$

↙ m阶l次连带Legendre函数

本征值 $\lambda_l = l(l+1)$, $l = m, m+1, m+2 \cdots$

$$P_l^m(x) = \frac{1}{2^l l!} (-1)^m (1-x^2)^{\frac{m}{2}} [(x^2-1)^l]^{(l+m)}$$

$m > l \Rightarrow 0 \Rightarrow m \le l$

磁量子数 角量子数

$$P_0^0 = 1, \quad P_1^0 = x \quad P_1^1 = -(1-x^2)^{\frac{1}{2}},$$

$$P_2^0 = \frac{1}{2}(3x^2-1), \quad P_2^1 = -3x(1-x^2)^{\frac{1}{2}}, \quad P_2^2 = 3(1-x^2)$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta = 0, \quad \Theta(0), \Theta(\pi) < \infty$$

$$\Rightarrow \begin{cases} \lambda_l = l(l+1) & l = m, m+1, m+2 \\ P_l^m(\cos\theta) \end{cases}$$

(2) 广义傅里叶级数

S-L理论: $\int_{-1}^{1} P_k^m(x) P_l^m(x)\,dx = 0, \quad k \neq l.$

$$\|P_l^m\|^2 = \int_{-1}^{1} [P_l^m(x)]^2 dx = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}$$

$[-1,1]$: $f(x) = \sum_{l=m}^{\infty} C_l P_l^m(x) \quad m = 1, 2, 3, \ldots$

$$C_l = \frac{1}{\|P_l^m\|^2}\int_{-1}^{1} f(x) P_l^m(x)\,dx$$

$[0,\pi]$: $f(\theta) = \sum_{l=m}^{\infty} C_l P_l^m(\cos\theta)$

$$C_l = \frac{1}{\|P_l^m\|^2}\int_0^{\pi} f(\theta) P_l^m(\cos\theta)\sin\theta\,d\theta$$

求模方计算:

$$\|P_l^m\|^2 = \int_{-1}^{1} (P_l^m(x))^2 dx = \int_{-1}^{1} (1-x^2)^m \left(P_l^{(m)}(x)\right)^2 dx$$

$$= \int_{-1}^{1} (1-x^2) P_l^{(m)} \left(P_l^{(m-1)}\right)' dx$$

$$= \underbrace{\left[(1-x^2) P_l^{(m)}\right] P_l^{(m-1)} \Big|_{-1}^{1}}_{=0} - \int_{-1}^{1} \left[(1-x^2)^m P_l^{(m)}\right]' P_l^{(m-1)} dx$$

m次分部积分

$$= (-1)^m \int_{-1}^{1} \left[(1-x^2)^m P_l^{(m)}\right]^{(m)} P_l\,dx$$

$$= (-1)^m \int_{-1}^{1} \frac{1}{2^l l!} \left[(1-x^2)^m \left((x^2-1)^l \right)^{(l+m)} \right]^{(m)} P_l \, dx$$

$$= \int_{-1}^{1} G_l(x) P_l(x) dx$$

$$G_l(x) = \frac{(-1)^m}{2^l \cdot l!} \left[(1-x^2)^m \left((x^2-1)^l \right)^{(l+m)} \right]^{(m)},$$

其次数为 $2m + 2l - (l+m) - m = l$，可写为 $G_l = a_l x^l + a_{l-1} x^{l-1} + \cdots + a_0$

$$= a_l \int_{-1}^{1} x^l P_l(x) dx \quad \leftarrow \int_{-1}^{1} x^k P_l(x) dx = 0 \ (k<l)$$，只有 $a_l x^l$ 项有贡献

$$= a_l \frac{2^{l+1} (l!)^2}{(2l+1)!} \quad \leftarrow \text{利用前几页结果}$$

而 $a_l x^l = \frac{(-1)^m}{2^l l!} \left[(-1)^m x^{2m} (x^{2l})^{(l+m)} \right]^{(m)}$ ← 取 G_l 中的最高次项

$$= \frac{1}{2^l l!} \left[x^{2m} \frac{(2l)!}{(l-m)!} x^{l-m} \right]^{(m)} = \frac{1}{2^l l!} \frac{(2l)!}{(l-m)!} \frac{(l+m)!}{l!} x^l$$

$$\Rightarrow a_l = \frac{(2l)!}{2^l (l!)^2} \frac{(l+m)!}{(l-m)!}$$

$$\Rightarrow \| P_l^m \|^2 = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!}$$

(3) 球谐函数. 球面调和函数 spherical harmonics

$$\nabla^2 u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial\phi^2} = 0$$

球内 Laplace 方程

$r<a$

$$u|_{\theta=0},\ u|_{\theta=\pi} < \infty$$

$$u|_{\phi=0} = u|_{\phi=2\pi},\ \left.\frac{\partial u}{\partial\phi}\right|_{\phi=0} = \left.\frac{\partial u}{\partial\phi}\right|_{\phi=2\pi}$$

$$u|_{r=a} = f(\theta,\phi),\quad u|_{r=0} < \infty$$

$$u(r,\theta,\phi) = R(r)S(\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left(\lambda - \frac{\mu}{\sin^2\theta}\right)\Theta = 0$$

$$\Phi'' + \mu\Phi = 0,\quad \Phi(0) = \Phi(2\pi)\cdots \Rightarrow \mu_m = m^2$$

$\cos m\phi$ $\sin m\phi$ $\quad m = 0, 1, 2\cdots$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta = 0$$

$$\lambda_l = l(l+1),\quad P_l^m(\cos\theta),\quad l = m, m+1\cdots$$

$$S_{lm1} = P_l^m(\cos\theta)\cos m\phi,\quad m = 0, 1, 2\cdots l$$

$$S_{lm2} = P_l^m(\cos\theta)\sin m\phi \quad m = 1, 2\cdots l$$

共 $2l+1$ 个

或 $e^{\pm im\phi}$

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-l(l+1)R=0 \xrightarrow{r=e^t} R=C_l r^l+\frac{D_l}{r^{l+1}}=C_l r^l$$

$$\Rightarrow u(r,\theta,\phi)=\sum_{l=0}^{\infty}\sum_{m=0}^{l} r^l P_l^m(\cos\theta)\left(A_{lm}\cos m\phi+B_{lm}\sin m\phi\right)$$

$$=\sum_{l=0}^{\infty}\sum_{m=0}^{l} r^l\left(A_{lm}S_{lm1}+B_{lm}S_{lm2}\right)$$

θ,ϕ 二重级数

仅当 $l=l'$, $m=m'$, $\alpha=\alpha'$ 时,

$$\int_0^{\pi}\int_0^{2\pi} S_{lm\alpha}(\theta,\phi)\,S_{l'm'\alpha'}(\theta,\phi)\sin\theta\,d\theta\,d\phi\neq 0$$

$\alpha=1,2$

$$\|S_{lm1}\|^2=\int_0^{\pi}\int_0^{2\pi}\left[P_l^m(\cos\theta)\right]^2\cos^2 m\phi\sin\theta\,d\theta\,d\phi=\frac{(l+m)!}{(l-m)!}\frac{2\pi}{2l+1}(1+\delta_{m0})$$

$$\|S_{lm2}\|^2=\int_0^{\pi}\int_0^{2\pi}\left[P_l^m(\cos\theta)\right]^2\sin^2 m\phi\sin\theta\,d\theta\,d\phi=\frac{(l+m)!}{(l-m)!}\frac{2\pi}{2l+1}$$

$$u|_{r=a}=f(\theta,\phi)=\sum_{l=0}^{\infty}\sum_{m=0}^{l} a^l\left(A_{lm}S_{lm1}+B_{lm}S_{lm2}\right)$$

$$\Rightarrow a^l A_{lm}=\frac{1}{\|S_{lm1}\|^2}\int_0^{\pi}\int_0^{2\pi} f(\theta,\phi)\,S_{lm1}(\theta,\phi)\sin\theta\,d\theta\,d\phi$$

$$a^l B_{lm}=\frac{1}{\|S_{lm2}\|^2}\int_0^{\pi}\int_0^{2\pi} f(\theta,\phi)\,S_{lm2}(\theta,\phi)\sin\theta\,d\theta\,d\phi$$

$$\Rightarrow A_{lm}=\cdots,\quad B_{lm}=\cdots$$

$m \le l$

$$f(\theta,\phi) = \sum_{l=0}^{\infty} \sum_{m=0}^{l} (\cdots) S_{lm1} + (\cdots) S_{lm2}$$ 称为 Laplace 级数.

θ, ϕ 二重级数

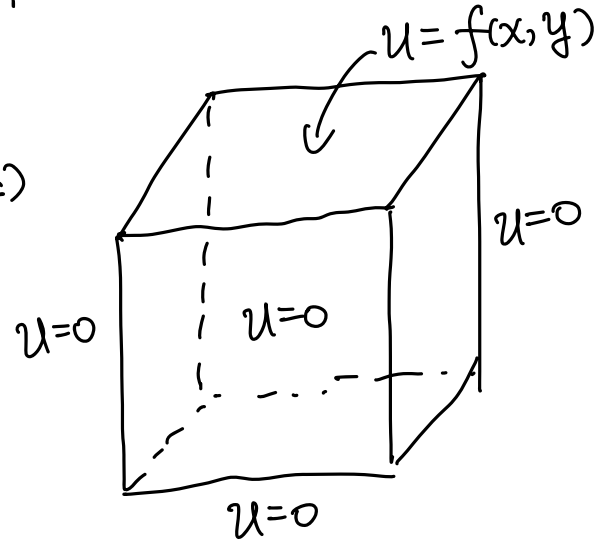
和矩形 $0<x<\pi$, $0<y<\pi$ 上的展开类似:

$$f(x,y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (\cdots) X_{nm}(x,y)$$

$$= \sin(nx)\sin(my)$$

球面和矩形的区别在于 $l \ge m$.

球内 Laplace 方程和立方体内 Laplace 方程类比:



上表面 $u = f(x,y)$

其余表面 $u=0$

分离变量法 $u(x,y,z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} R_{nm}(z) X_{nm}(x,y)$

(θ,ϕ)

类比 $R(r)$ r S_{lm1}, S_{lm2}